

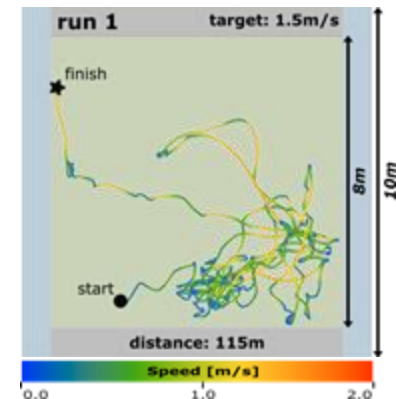
A Sim-to-Real Deep Learning-based Framework for Autonomous Nano-drone Racing

Lorenzo Lamberti, Elia Cereda, Gabriele Abbate, Lorenzo Bellone, Victor Kartsch, Michał Barciś, Agata Barciś, Alessandro Giusti, Francesco Conti, Daniele Palossi

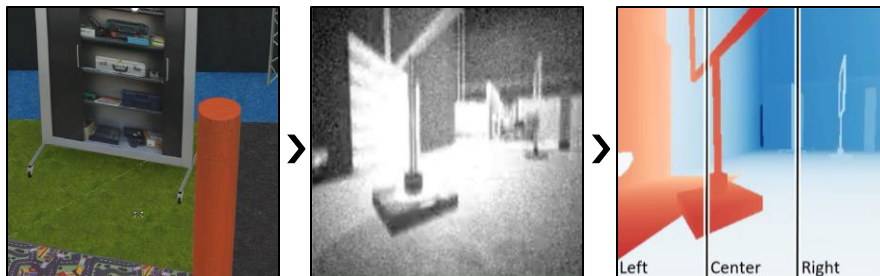
“Nanocopter AI Challenge” @ IMAV’22

Autonomous vision-based navigation in an unknown flight arena, with static and dynamic obstacles.

**We won by flying 115m in 5’
without any collisions.**



Sim-to-real mitigation strategy

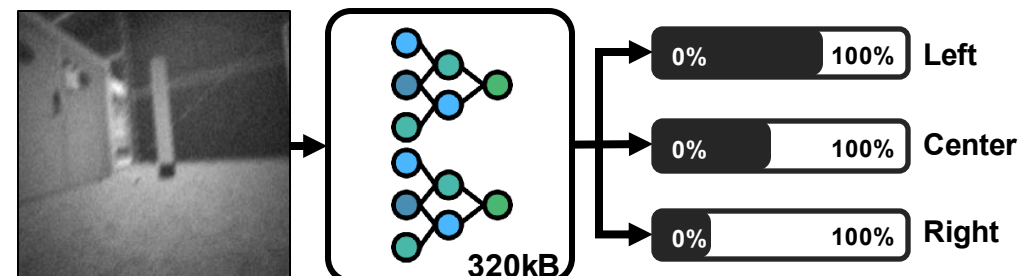


Simulator

Augmentation

Depth labels

CNN → fully onboard execution @ 30FPS



Camera

CNN

Collision probability